Exam 2025

Advanced Methods in Applied Statistics

Chamilla Terp
UCPH ID: bmx626

March 28, 2025

I, Chamilla Terp, expressly vow to uphold my scientific, academic, and moral integrity by working individually on this exam and soliciting no direct external help or assistance.

Problem 1

Solution - Problem 1a

In order to identify the function from which the data[1] in columns 1, 2, and 3 have been generated, I first plot the data from those three columns as seen in Figure 1. For simplicity I have called all the variables (i.e. the data) x.

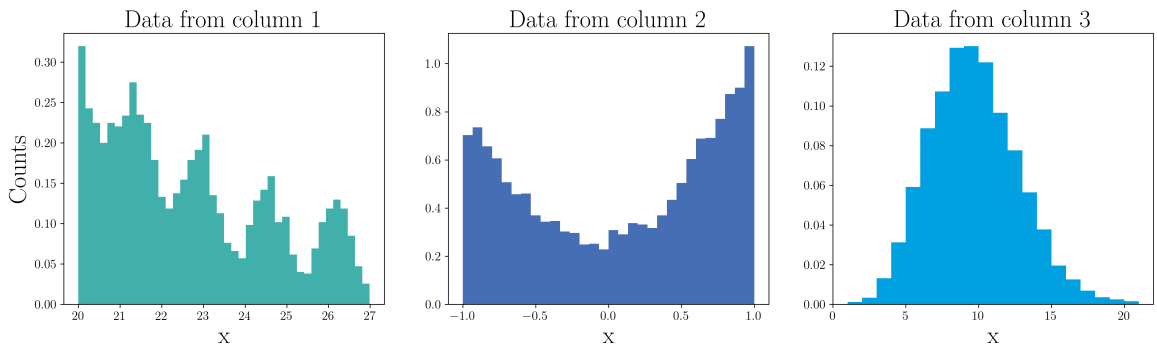

Figure 1: Plots showing the data from columns 1, 2, and 3.

All functions used is using the following ranges: $-10 \leq a \leq 10$, $-10 \leq b \leq 10$, and $4000 \leq c \leq 8000$. Below I describe the process for fitting the data in the three columns. The best-fit parameters along with reduced χ^2- and p-values are shown in Table 1.

Column 1

First I look at the data in column 1 (left-most plot in Figure 1). Based on looking at the data in the `.txt` file, I see that it consists of *floats*, which eliminates all of the discrete functions. Furthermore, both by visually inspecting the plot and the data, it seems like there is some kind of wave-like behavior along with some type of decrease in the overall number of counts as x becomes bigger. Assuming that this decrease is exponential, I test the following function:

$$f(x) \propto sin(ax) + ce^{bx} + 1. \tag{1}$$

Eq. 1 is normalized over the range $20 \leq x \leq 27$ using standard trapezoidal summation. Since the data visually shows multiple local minima and maxima, I choose to use a Bayesian approach for fitting the data, in order to account for the complexity in the data. For the Bayesian approach I utilize nested sampling through the package `UltraNest`[2]. The priors used are flat and bounded within the defined parameter ranges. I sample 1.000 points from the priors, which results in the corner plot seen in Figure 2. As seen in the corner plot both a and b seems to be well constrained, with a being slightly better constrained as seen by its symmetric distribution. The parameter c on the other hand, is not nearly as well constrained as a and b. The posterior distribution is both broad and asymmetric, indicating significant uncertainty.

Column 2

Looking at the data in column 2, I can rule out all of the discrete functions, as the data is once again *floats*. Visually (middle plot in Figure 1) it seems like the data could be drawn from a combination of a linear and parabolic function. Therefore, I choose to fit the data with the following function:

$$f(x) \propto 1 + ax + bx^2. \tag{2}$$

From eq. 2, I can quite easily compute the analytical form of the normalization constant (given the range $x \in [-1, 1]$) as $\int_{-1}^{1} 1 + ax + bx^2 dx = \frac{2}{3}(b + 3)$. To fit the data with the normalized version of eq. 2, I use the maximum likelihood approach through the package `Minuit`. For this data, I use 100 bins when computing the reduced χ^2- and p-value.

[1] Link: https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2025/data/Exam_2025_Prob1.txt.

[2] Following the procedures described in the package's documentation: https://johannesbuchner.github.io/UltraNest/index.html [Accessed on the 27th of March 2025].

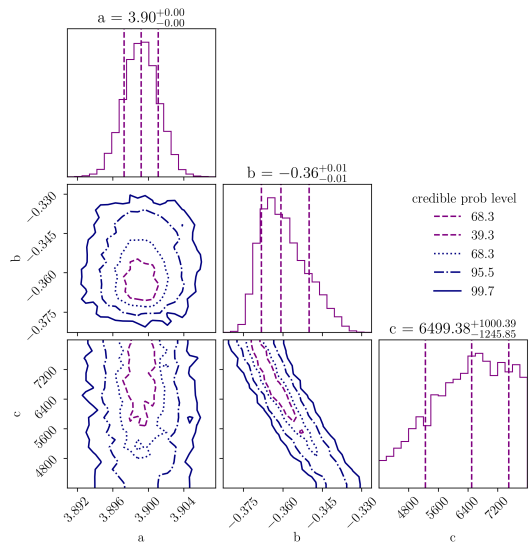

Figure 2: Corner plot produced using `UltraNest` with 1.000 sampling points. The corner plot show the distribution and correlation of the parameters a, b, and c.

Column 3

Opposite the data in columns 1 and 2, the data in column 3 is *integers*, which eliminates the continuous functions. When inspecting the data visually (right-most plot in Figure 1), I note that shape further eliminates the logarithmic function. This leaves a Poisson and binomial function (both are probability mass functions (PMFs)):

$$\text{Binomial: } f(k) = \binom{n}{k} p^k (1-p)^{n-k}, \tag{3}$$

$$\text{Poisson: } f(k) = \frac{\lambda^k e^{-\lambda}}{k!}. \tag{4}$$

Since eqs. 3 and 4 are PMFs, they are already normalized and I do not need to do anything else. I choose to fit both function to the data, using the same approach as for the data in column 2. However, the number of bins used is instead 30.

Collection of results

The best-fit parameters, reduced χ^2- and p-values for the fits described above can be seen in Table 1.

	Column 1	**Column 2**	**Column 3**	
	Eq. 1	Eq. 2	Eq. 3	Eq. 4
a	3.899 ± 0.002	0.47 ± 0.05	-	-
b	$-0.359^{+0.009}_{-0.009}$	2.58 ± 0.16	-	-
c	$(6.40^{+1.12}_{-1.13}) \times 10^3$	-	-	-
n	-	-	$(11.0 \pm nan) \times 10^3$	-
p	-	-	$0.001 \pm nan$	-
λ	-	-	-	9.18 ± 0.04
χ^2_{red}	0.80	0.84	10.30	0.55
p-value	0.93	0.88	0.00	0.95

Table 1: Best-fit parameters, reduced χ^2- and p-values for each fit.

All of the reduced χ^2-values (except for the binomial fit) are all below one, suggesting that the fits are almost *too* good - which means that the Poisson errors used to compute the reduced χ^2-values are being overestimated (i.e. too big). It becomes very obvious for the discrete Poisson function (eq. 4), where the number of bins used to compute the values are much lower than for the continuous functions (eqs. 1 and 2). Furthermore, the p-values between 0.88 and 0.95 (excluding

the $p = 0$ value for the binomial function) indicate that I cannot reject the hypotheses that the data are sampled from the functions/distributions. So, based on these statistics I can only conclude that I cannot reject the hypotheses that the data from column 1 and 2 are sampled from the functions presented - but I also cannot reject the hypotheses that the data is drawn from different functions. However, I can conclude that it seems like the data in column 3 is not drawn from the binomial function. But I cannot favor any functions with significant confidence based on these statistics and results.

Solution - Problem 1b

The data from the columns 1, 2, and 3 along with the best-fit function(s) are shown in Figure 3.

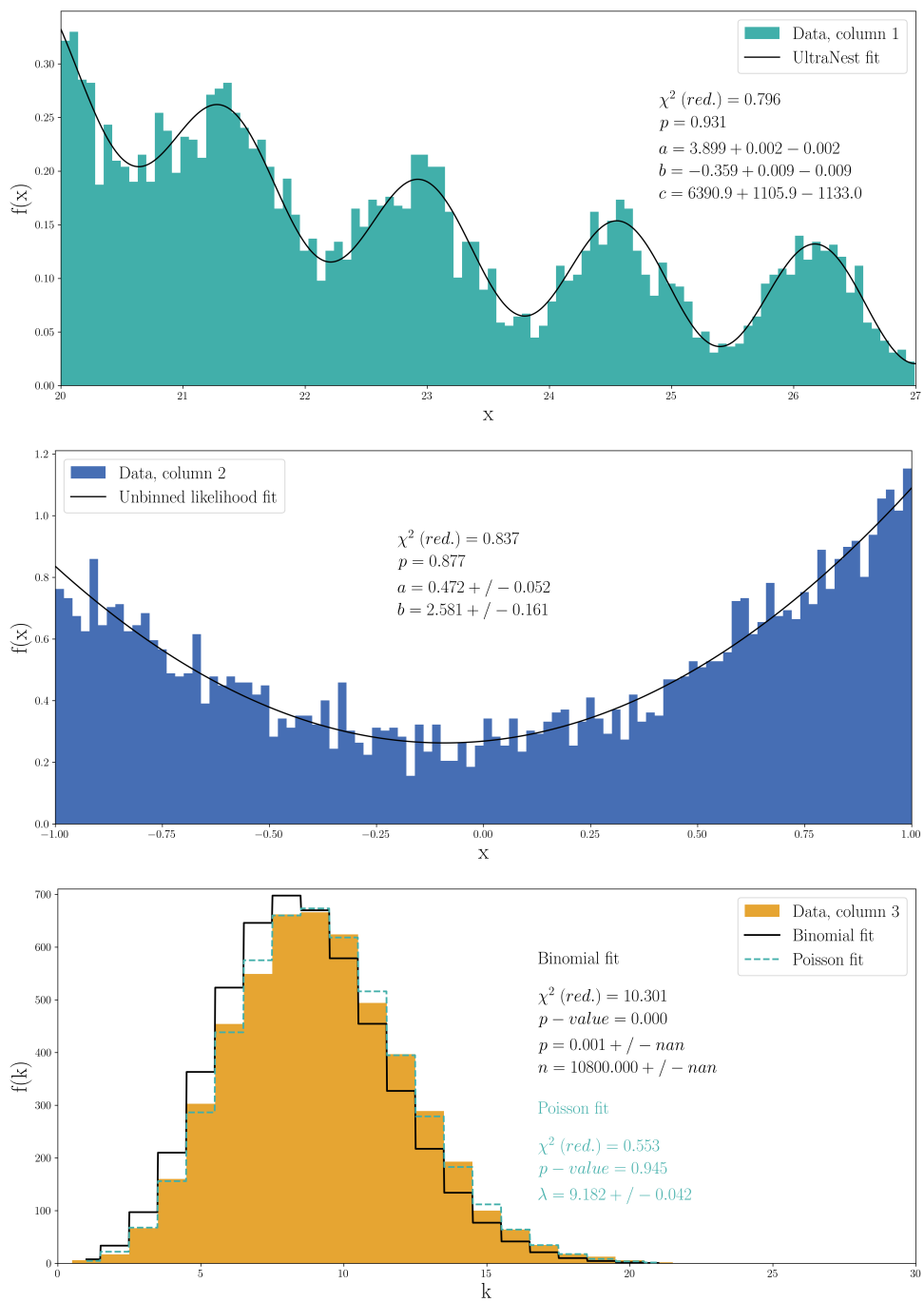

Figure 3: Data from the three columns along with the best-fit function(s).

Problem 2

Solution - Problem 2a

To quantify if the data[3] provided in this problem is spherically isotropically distributed, I compare the data to spherically isotropic Monte Carlo generated data. More specifically, the data is sampled from a spherically isotropic distribution (i.e. uniform in the azimuth angle from 0 to 2π, and uniform in the cos(zenith angle) from 1 to -1) where the size of the generated data is equal to that of the real data to ensure a fair comparison. In Figure 4 a Mollweide projection of the real data and the generated data can be seen.

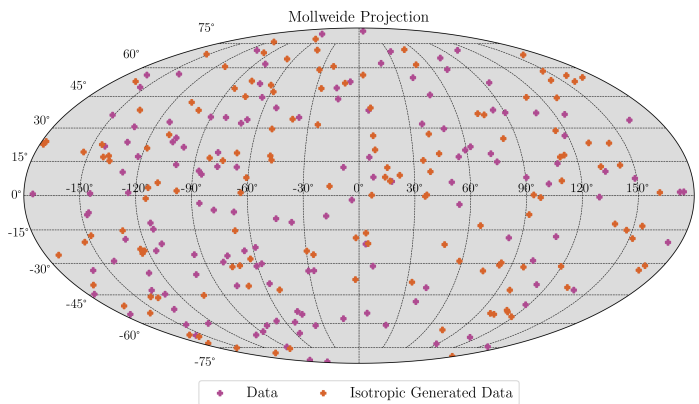

Figure 4: Mollweide projection of the data (magenta) and the generated data (orange).

To test whether the data is spherically and isotropically distributed, I evaluate the performance of both the real and generated datasets using a series of pseudo-experiments. In each experiment, I apply a Kolmogorov–Smirnov (KS) test along with the cumulative two-point autocorrelation function ($\mathcal{C}(cos(\phi))$) to identify potential clustering patterns. This methodology allows me to compute the probability of rejecting the null hypothesis that the data is spherically isotropic.

The two-point autocorrelation function is given by[4]:

$$\mathcal{C}(\mathbf{n}_i, \phi) = \frac{2}{N_{tot}(N_{tot}-1)} \sum_{i=1}^{N_{tot}} \sum_{j=1}^{i-1} \Theta(cos(\phi_{ij})) - cos(\phi)), \tag{5}$$

where $cos(\phi_{ij}) = \mathbf{n}_i \cdot \mathbf{n}_j$ is the angular distance between two points. For perfect isotropic data, the cumulative two-point autocorrelation function becomes:

$$\mathcal{C}_{isotropic}(\mathbf{n}_i, \phi) = \frac{1}{2}(1 - cos(\phi)). \tag{6}$$

Because both eqs. 5 and 6 are cumulative, they can be used in the KS test, with the following test statistic:

$$KS(\{\mathbf{n}_i\}) = \mathbf{sup}_\phi |\mathcal{C}(\mathbf{n}_i, \phi) - \mathcal{C}_{isotropic}(\mathbf{n}_i, \phi)|. \tag{7}$$

The results of this analysis are presented in Figure 5.

The KS-test on the provided data yields a KS-test statistics value of 0.0221. However, to assess whether this is significant enough to reject the null hypothesis, additional statistical context is required. The next step in the process, is thus to perform the "experiment" many times. I therefore run 10.000 pseudo experiments, where the KS-test statistics value is calculated as above for each experiment. The distribution of KS-values is shown in Figure 6. Based on this distribution, I calculate the p-value as the fraction of sampled KS-values that are greater than or equal to the KS-value obtained from the real data. This yields a p-value of $p = 0.0026$, indicating that the null hypothesis can be rejected with over 99% confidence - meaning that the data does not seem to be spherically isotropic.

[3] Link: https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2025/data/Exam_2025_Problem2.txt.

[4] Can be found in these lecture notes: https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2025/Lecture_AhlersKoskinen2025.pdf, slide 28-34 [Accessed on the 27th of March 2025].

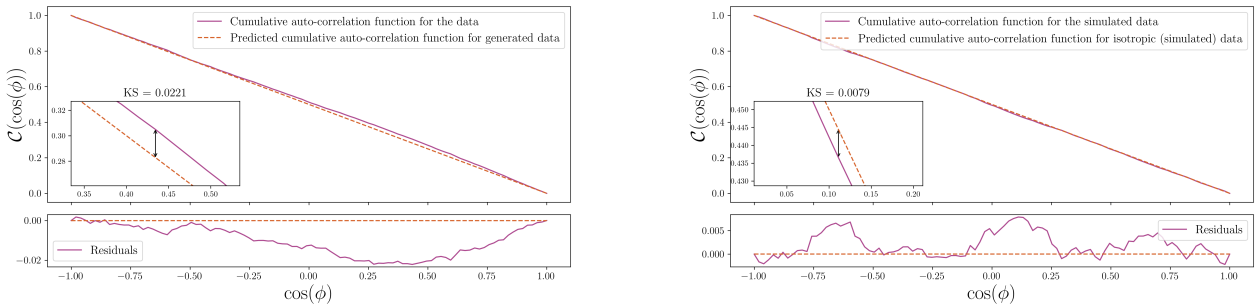

Figure 5: Results of the KS-test of the cumulative two-point auto-correlation function for the provided data (left) and the generated data (right).

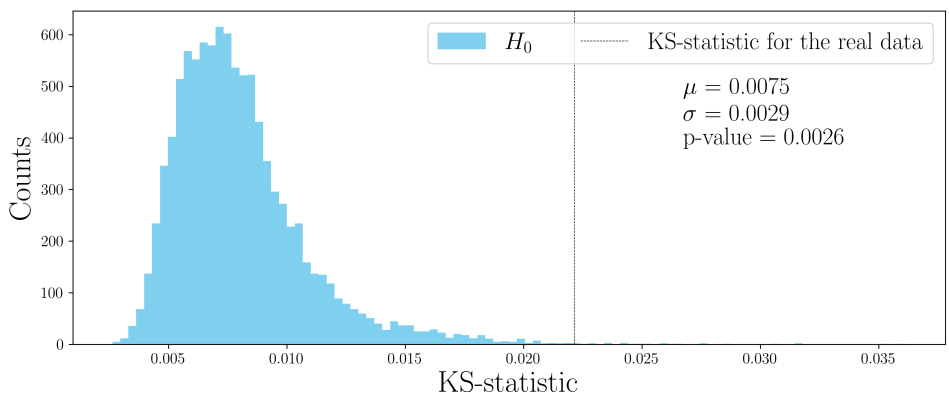

Figure 6: KS-test statistics values distribution based on 10.000 pseudo experiments with Monte Carlo generated spherically isotropic distributed data. The KS-value from the real data set is marked with a vertical dashed black line. On the plot μ refers to the median of the KS-values distribution and σ is the standard deviation of the distribution.

Solution - Problem 2b

In order to compare the two alternative hypotheses, H_A and H_B, I follow the procedure described above - now the generation of Monte Carlo samples is just modified to fit the new hypotheses. The resulting distribution of KS test statistics for both hypotheses, alongside with the distribution from Problem 2a, are shown in Figure 7. When computing the p-values for the comparisons with the null hypothesis, I use the median values of the KS distribution generated under H_A and H_B. The p-values are presented in Table 2 below.

	H_A vs. H_0	H_B vs. H_0
p-value	0.0102	0.0872

Table 2: Calculated p-values of H_0 vs. the alternative hypotheses.

Based on the p-values shown in Table 2, a 95% significance level leads to the conclusion that hypothesis B is somewhat consistent with the null hypothesis while hypothesis A is not. However, this does not say anything about whether the new hypothesis fit the data. In order to asses that, the p-values for the data vs. each of the alternative hypothesis is computed and can be seen in Table 3. Based on these values and a 95% significance level, neither of the alternative hypotheses can be rejected. However, the data appears most consistent with hypothesis A, which aligns with the earlier observation that hypothesis A is least consistent with the null hypothesis.

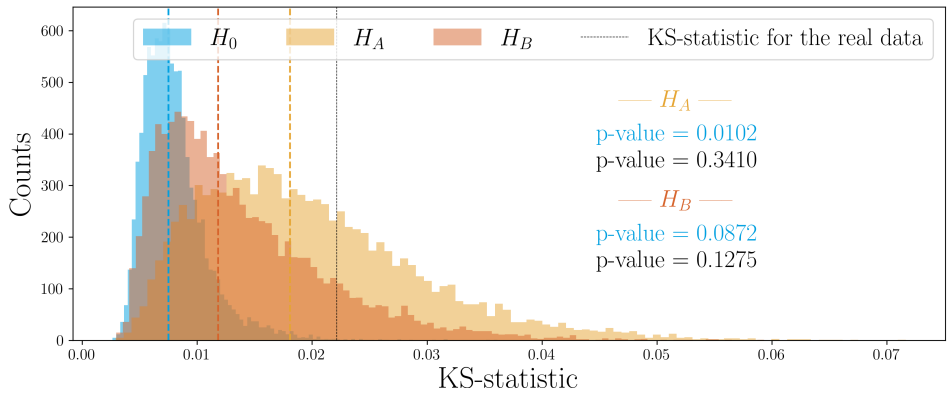

Figure 7: KS-test statistics values distribution based on 10.000 pseudo experiments with Monte Carlo generated data based on three different hypotheses. The KS-value from the real data set is marked with a vertical dashed black line. Colored dashed lines show the median values of each distribution.

	Data vs. H_0	Data vs. H_A	Data vs. H_B
p-value	0.0026	0.3410	0.1275

Table 3: Calculated p-values for the data vs. the null hypothesis, as well as the alternative hypotheses.

Problem 3

Solution - Problem 3a

To estimate the expected number of times a student, who has a Mayo & Cheese sandwich for 200 meals, also has chocolate I first model the given ingredient ratio. The ingredient ratio for the Mayo & Cheese sandwich is given as a beta distribution, $F_{beta(\alpha,\beta)}$, with $\alpha = 7.7$ and $\beta = 30$. The fraction of mayonnaise in a sandwich follows this beta distribution, and thus the cheese fraction is simply 1 minus the mayonnaise distribution. According to the information given, a student only eats chocolate if the cheese faction exceeds 0.83, i.e., if the mayonnaise fraction is less than 0.17. This probability can be computed using the cumulative distribution function (CDF) of the beta distribution:

$$P(\text{mayonnaise} < 0.17) = F_{\text{beta}(7.7,30)}(0.17). \tag{8}$$

This value is found to be ~ 0.32. Now, multiplying this probability with the number of days, i.e., 200 days, yields the expected number of times a student also eats chocolate. So, $200 \times 0.32 \approx 64.44$ times. A plot showing the two distributions can be seen in Figure 8.

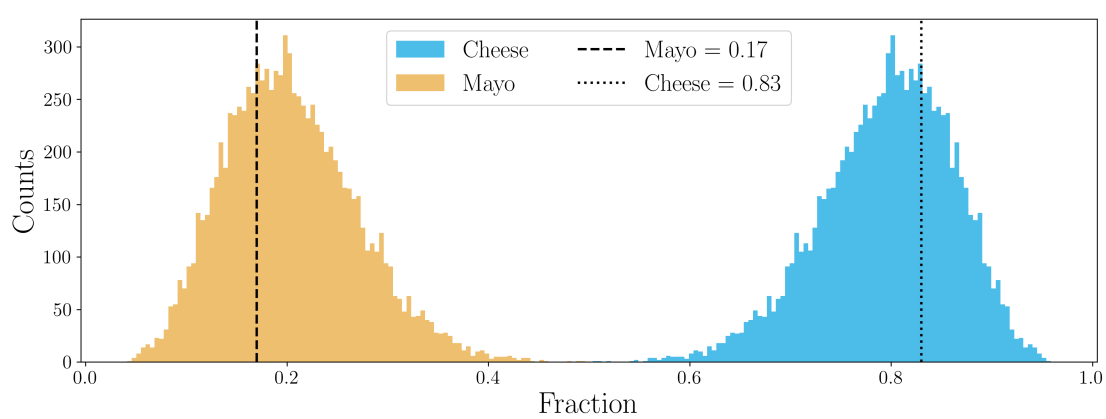

Figure 8: Beta distributions for the ingredient composition of mayonnaise and cheese.

Solution - Problem 3b

Now to estimate the expected number of days in a year, where a student eats chocolate, I implement a Monte Carlo simulation based on the given meal probabilities and chocolate conditions. Each day, the student selects one of five possible meals at random, with the probabilities and special rules shown in Table 4.

Meal	Probability	Special rule
Mayo & Cheese sandwich	0.20	Chocolate if cheese fraction ≥ 0.83
Frozen Pizza	0.25	Always chocolate
Frozen Cornish Pasties	0.10	Never chocolate
Rice & Eggs	0.33	Chocolate if SHU ≥ 21.4
Spaghetti Bolognese	0.12	Chocolate if SHU ≥ 8.2

Table 4: Meal probabilities and special rules. SHU is the Scoville Heat Unit.

The simulation was run 10.000 times (years), where each run simulate a full year of 365 meals. In each simulation, I count the number of days a student eats chocolate according to the rules above. The result of the simulation is shown in Figure 9. As see in Figure 9, I find that based on the probabilities of a student choosing each meal and the special rules, a student would eat chocolate ~ 134 days a year with an inter-quartile range of 13 days.

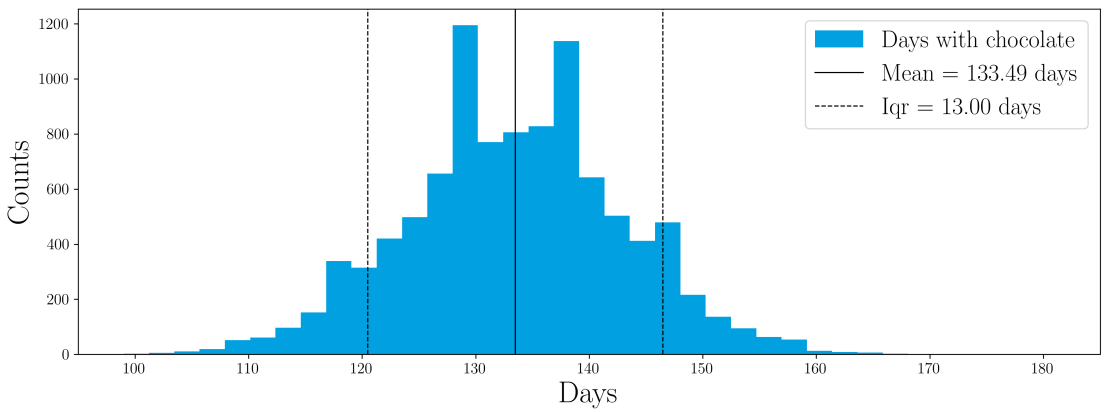

Figure 9: Histogram of results from the simulation. The mean value is marked with a black line as well as the inter-quartile range with dashed black lines.

Solution - Problem 3c

For the last part of this problem, I want to estimate the probability that a student will eat Spaghetti Bolognese for three consecutive days within a single year. To do this, I once again use a Monte Carlo simulation. In each run of the simulation, a sequence of 365 meals is generated by randomly sampling from a simple Bernoulli distribution with success probability equal to the probability of choosing Spaghetti Bolognese. Each element in the simulated sequence is `True` (1) if the meal is Spaghetti Bolognese and `False` (0) if not. I then check is there exists any 3-day window, where all three meals are Spaghetti Bolognese (thus in principle, a student could eat Spaghetti Bolognese for more than 3 days. But the criteria is just *at least* 3 days). This process is repeated over 10.000 years, and I count how many of those contain at least one 3-day Spaghetti Bolognese streak.

Based on the simulation, I get $P(3 \text{ consecutive Spaghetti Bolognese dags in a year}) \approx 0.438$, which means that the chance of a student eating Spaghetti Bolognese for three consecutive days at least once is $\sim 44\%$. So, although the daily probability of eating Spaghetti Bolognese is relatively low (12%), the large number of trials (i.e. the 365 days in a year) make such 3-days streaks more likely than might be intuitively expected. The result illustrates how even rare daily events can accumulate into noticeable patterns over time.

Problem 4

Solution - Problem 4a

To test the two different hypotheses, I perform two different unbinned likelihood fits (following the same procedure as in Problem 1) on the provided data[5].

Double Gaussian, Hypothesis A (H_A)

In this approach, I fit two separate Gaussian probability density functions (PDFs) to the data and combine them to form the total density function. During the fitting process, I ensure that the relative sizes of the individual Gaussians are preserved and that the combined PDF is properly normalized. The normalization is performed using trapezoidal summation. The resulting fit is shown in the left plot in Figure 10.

Triple Gaussian, Hypothesis B (H_B)

To evaluate the triple Gaussian hypothesis, I retain the previously fitted Gaussian for the left peak and model the right peak using a double Gaussian. This approach is motivated by the proximity of the two right-hand peaks, which makes it difficult to fit them with two entirely independent Gaussians. As before, I preserve the relative sizes of the individual components and normalize the combined PDF. The resulting fit is shown in the right plot in Figure 10.

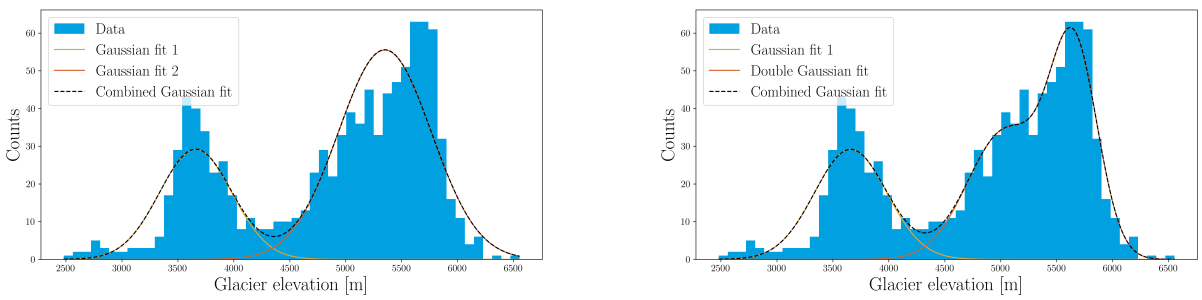

Figure 10: The two separate likelihood fits are shown with the yellow and orange lines, while the total PDF is shown as a dashed black line. Left plot show the double Gaussian and the right show the triple Gaussian.

Solution - Problem 4b

In order to statistically compare the two hypotheses, I calculate the reduced χ^2- and p-values. For each hypothesis I use 51 bins. The results can be seen in Table 5.

	Double Gaussian, H_A	Triple Gaussian, H_B
χ^2_{red}	4.62	4.41
p-value	0	0

Table 5: Calculated χ^2_{red}- and p-values for each hypothesis.

As seen in Table 5, the reduced χ^2-value is slightly lower for the triple Gaussian compared to the double Gaussian, which could be expected since it has more parameters and thus fewer degrees of freedom. However, when considering both the reduced χ^2 and the p-values, neither hypothesis appears to be a good fit for the data. In particular, the single Gaussian used to model the left-most peak struggles to fully capture the shape of the data. This suggests that an even more complex probability density function may be necessary to correctly describe the full distribution.

Before definitively rejecting both hypotheses based on these results, I perform an additional test. Instead of just computing one p-value, I perform 10.000 pseudo experiments where data from each PDF is sampled and the p-values are computed using the KS-test. For each experiment I sample the same number of data points as given in the original data. From this

[5]Link: https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2025/data/Exam_2025_Prob4.txt.

KS-test, one would expect the distribution of p-values to be uniform, if there is an agreement between the model and the data. The result of this simulation can be seen in Figure 11.

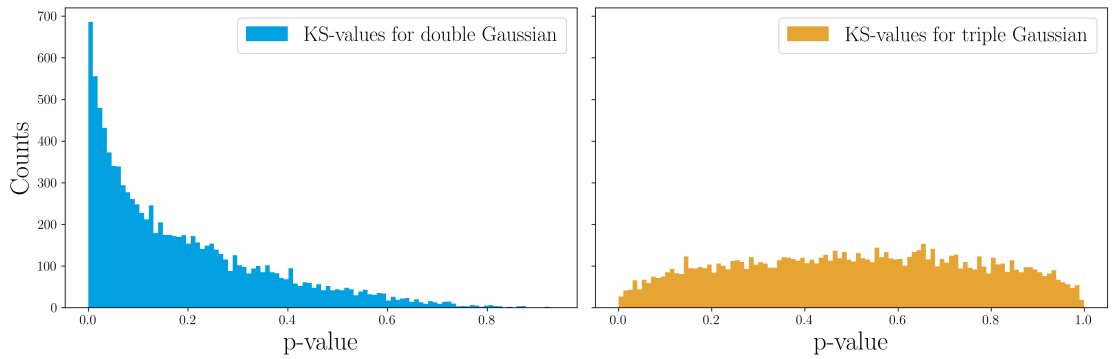

Figure 11: p-value distributions from KS-test of 10.000 Monte Carlo generated data sets sampled from the two hypotheses.

Based on the distributions shown in Figure 11, the double Gaussian model yields a distribution of p-values that is skewed toward lower values, indicating a poor agreement between the model and the data. The distribution of p-values for the triple Gaussian looks much more uniform, apart from small deviations. Based on this two distributions and combined with the results presented in Table 5, I would reject the double Gaussian hypothesis. However, I would still not fully trust the triple Gaussian hypothesis and thus one should still search for alternative models/hypotheses.

Problem 5

To determine the critical threshold for the test-statistics (the provided data[6]) on a one-sided p-value of 4.55%, I use the data - i.e. the empirical bootstrap distribution of 3.000 test-statistics samples. First the data is sorted, and then the 94.45th percentile is found using the Python function `np.percentile`. This corresponds to the cutoff above which only 4.55% of the values lie. Doing this, yields a bootstrap critical value of ~ 15.

For comparison, the theoretical critical value for a χ^2 distribution with 5 degrees of freedom is calculated. This yields $\chi^2_{0.9445,k=5} \approx 10.80$ or ~ 11. The bootstrap critical value is substantially higher than the theoretical value, which indicates that the actual distribution of the test-statistics is more *right-skewed* than the $\chi^2_{k=5}$ distribution. This can also be seen in Figure 12, where a histogram of the bootstrap samples is shown along with the $\chi^2_{k=5}$ PDF. Visually, the bootstrap distribution is seen to be shifted to the right and is also flatter, with heavier tails. Therefore, it seems like the bootstrap data does not follow a $\chi^2_{k=5}$ distribution, and thus using the theoretical critical value would underestimate the true critical value.

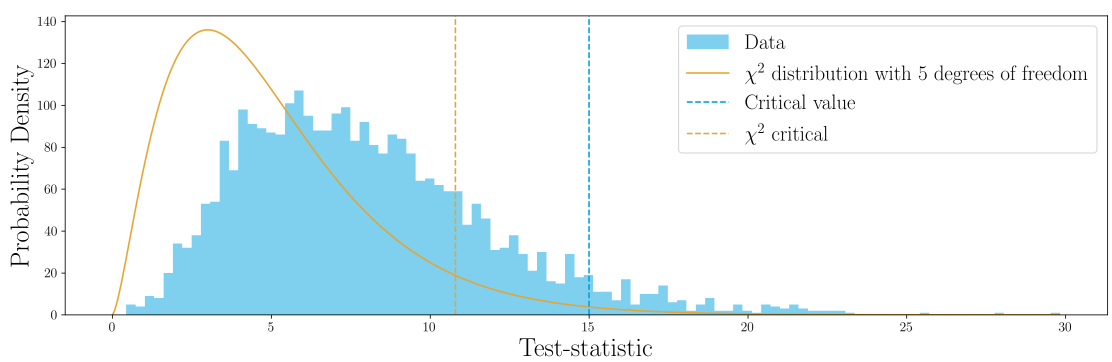

Figure 12: The bootstrap distribution shown alongside the $\chi^2_{k=5}$ distribution. The two vertical lines show the critical values for the bootstrap and $\chi^2_{k=5}$ distribution, respectively.

[6]Link: https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2025/data/Exam_2025_Problem5.txt.